

Two players share one pile of counters and take turns. On each turn a player takes away 1, 3, or 4 counters, and nobody may take more counters than the pile has. Whoever takes the last counter wins.

A pile is a **losing pile** if the player whose turn it is will lose when the other player plays carefully. A pile is a **winning pile** if the player whose turn it is can win, whatever the other player does.

**Problem 1:** Play the game several times with a pile of 6 counters, taking turns going first. Which player can always win from 6, the first player or the second player? Write how that player wins.

**Problem 2:** Write W under each winning pile and L under each losing pile.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|
|   |   |   |   |   |   |   |   |   |    |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|----|----|----|----|----|----|----|----|----|----|
|    |    |    |    |    |    |    |    |    |    |

**Problem 3:** A player claims: “If the pile has at least 4 counters and you can win, then taking 4 counters is a way to win.” Is the claim correct?

**Problem 4:** A player claims: “With a pile of 8 counters, the second player can always win by taking the same number of counters the first player just took.” Is the claim correct?

**Problem 5:** This table shows a game that started with 16 counters. Find every turn where the player could have taken a different number of counters and then been sure to win.

| turn | player | pile before | taken | pile after |
|------|--------|-------------|-------|------------|
| 1    | first  | 16          | 1     | 15         |
| 2    | second | 15          | 4     | 11         |
| 3    | first  | 11          | 1     | 10         |
| 4    | second | 10          | 3     | 7          |
| 5    | first  | 7           | 4     | 3          |
| 6    | second | 3           | 3     | 0          |

**Problem 6:** With a pile of 14 counters, which player can always win? Write a plan for that player that works against every move the other player might make.

**Problem 7:** For each pile, write W in its box if it is a winning pile and L if it is a losing pile. Write how you know the answer for 100.

|    |    |    |    |     |
|----|----|----|----|-----|
| 23 | 28 | 33 | 50 | 100 |
|    |    |    |    |     |

**Problem 8:** Play with a different rule: on each turn a player takes away 1, 3, or 5 counters. Whoever takes the last counter wins. Write W under each winning pile and L under each losing pile. Then describe all the losing piles, including piles larger than 20.

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|   |   |   |   |   |   |   |   |   |    |

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|    |    |    |    |    |    |    |    |    |    |

**Problem 9:** Now there are two piles. On each turn a player chooses one pile and takes 1, 3, or 4 counters from that pile only. Whoever takes the last counter wins. For each start, circle the player who can always win.

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| <div><div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div>7</div></div>            | <div><div><div><div></div><div></div></div></div><div>2</div></div>                                                        | <div>first player</div> | <div>second player</div> |
| <div><div><div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div>6</div></div>                       | <div><div><div><div></div></div></div><div>1</div></div>                                                                   | <div>first player</div> | <div>second player</div> |
| <div><div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div>9</div></div> | <div><div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div>8</div></div> | <div>first player</div> | <div>second player</div> |
| <div><div><div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div>6</div></div>                       | <div><div><div><div></div><div></div><div></div><div></div></div></div><div>4</div></div>                                  | <div>first player</div> | <div>second player</div> |